

CAPSTONE PROJECT · AI ENGINEER LAUNCHPAD

ResearchMate

Multi-Paper Analyzer

A multi-agent AI research assistant that reads, compares, and questions academic papers for you — powered by Retrieval-Augmented Generation.

FAISS

Sentence-Transformers

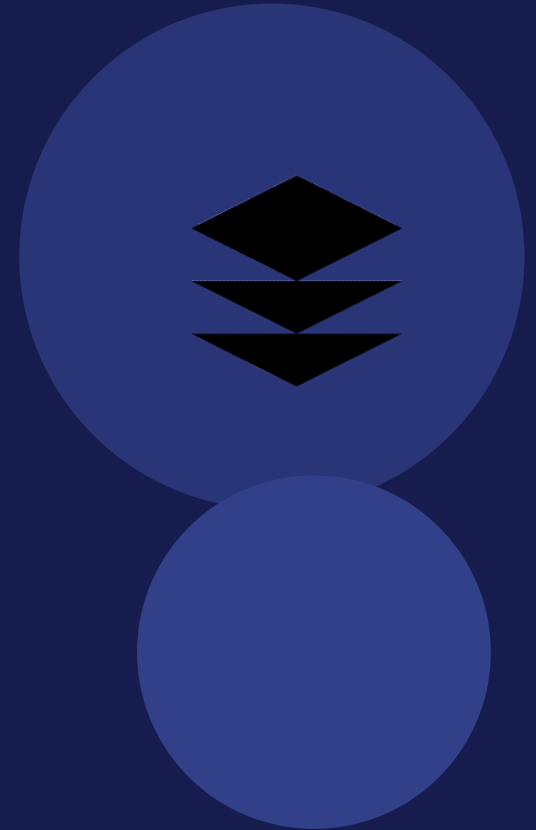
Groq LLM

LangChain

Streamlit

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The Research Bottleneck

Reviewing academic literature means reading dozens of long, dense PDFs — then manually comparing methods and hunting for gaps. Generic chatbots summarize text but aren't built around how researchers actually work.



Information Overload

Manually reading & re-reading long papers to answer one specific question.



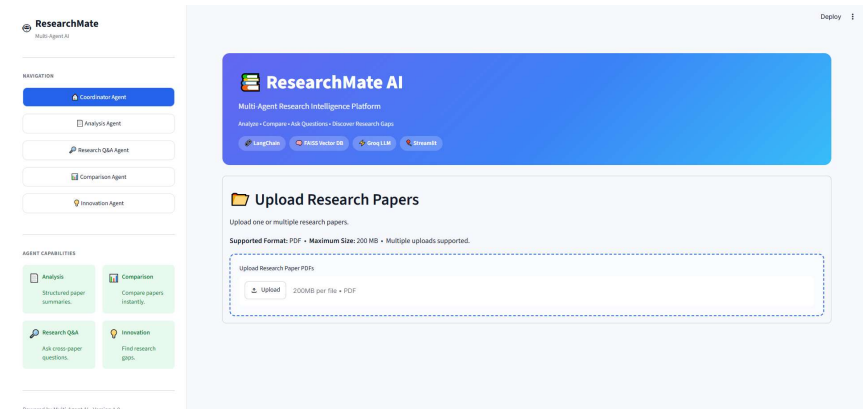
Fragmented Analysis

Comparing methods, datasets, and results across papers is done ad-hoc.



Hidden Research Gaps

Spotting what a set of papers collectively fails to address takes deep expertise.



The gap between information available and information accessible is the core problem ResearchMate solves.

Introducing ResearchMate

A Streamlit application where a user uploads one or more research PDFs and interacts with them through five specialized AI agents — instead of one generic chatbot.

1

Upload Papers

Drop up to 6 PDF research papers into the app.

2

Ask Anything

Chat naturally — the Coordinator Agent detects your intent.

3

Get Grounded Answers

Specialist agents respond with citations and structured output.

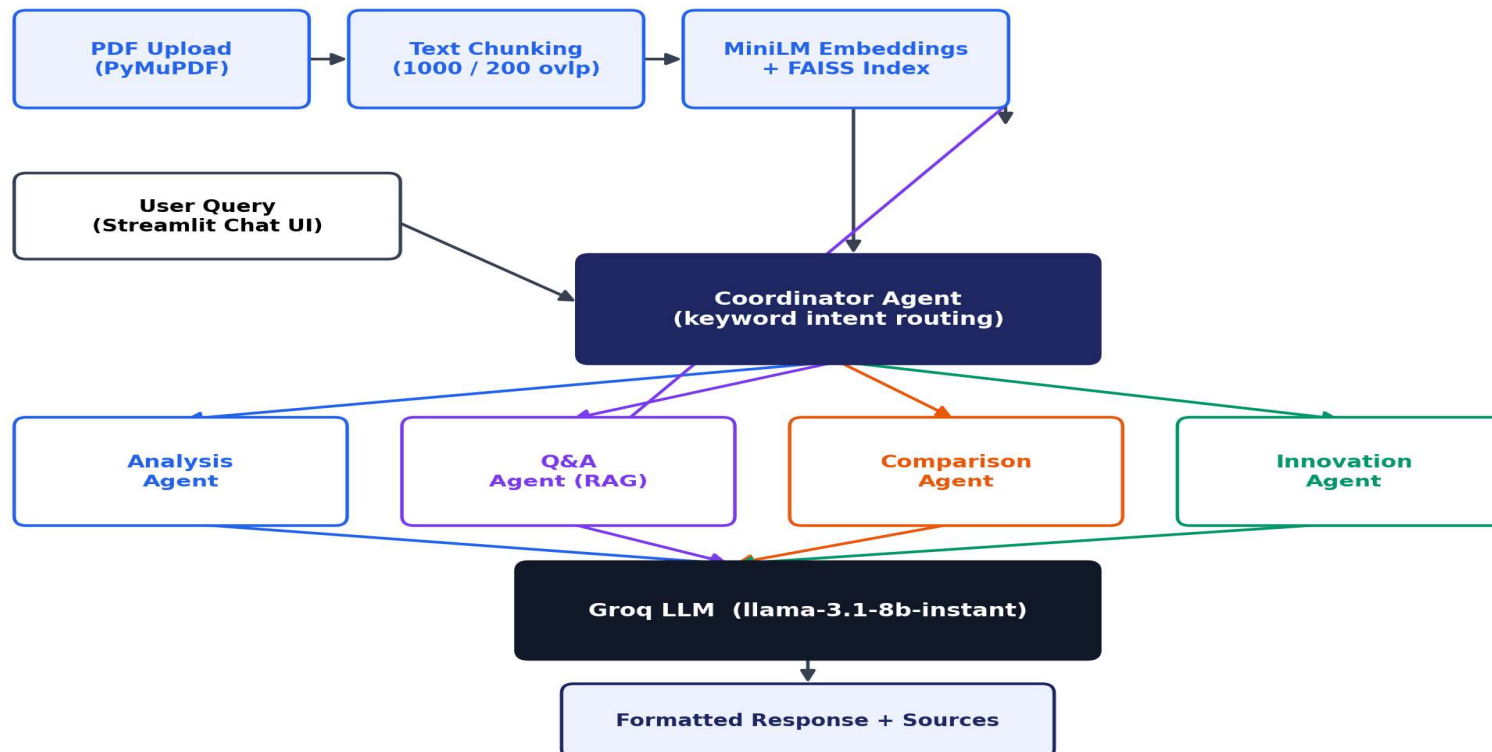
Core Idea

Instead of one monolithic chatbot, ResearchMate mirrors how a real research team divides work: a Coordinator Agent reads the user's intent and routes the query to the right specialist — Analysis, Q&A, Comparison, or Innovation.

System Architecture

A classic Retrieval-Augmented Generation pipeline wrapped in a multi-agent, intent-routing system.

ResearchMate: Multi-Agent RAG Architecture



A Team of Five Specialized Agents

Each agent owns one research sub-task — mirroring how a real literature-review team divides labour.



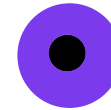
Coordinator Agent

Keyword-based intent detection. Routes each query to the right specialist(s) — no LLM call itself.



Research Analysis Agent

Structured 7-point paper review: problem, approach, methodology, contributions, limitations.



Research Q&A Agent

True RAG — FAISS semantic search over chunks, with explicit chunk-level source citations.



Comparative Intelligence Agent

Compares 2+ papers automatically or against a specific question; outputs a markdown table.

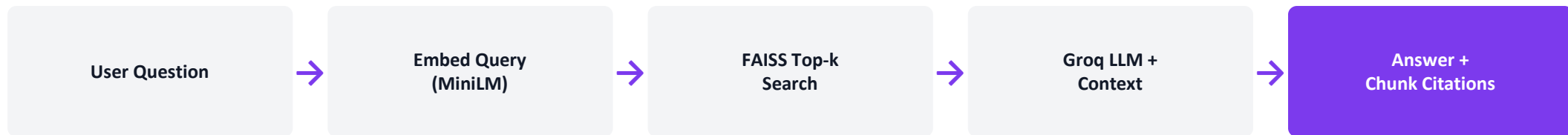


Research Innovation Agent

Surfaces research gaps, limitations, and proposes novel hybrid research directions.

Deep Dive: Research Q&A Agent

The only agent with genuine chunk-level retrieval — grounding every answer in the paper's actual text.



```
top_k = 3
if broad_question(question):
    top_k = min(10, len(chunks))

chunks, idx = search_vector_store(
    question, vector_store,
    embedding_model, chunks, top_k
)

answer = generate_answer(question, context)
answer += f'Sources: {sources}'
```

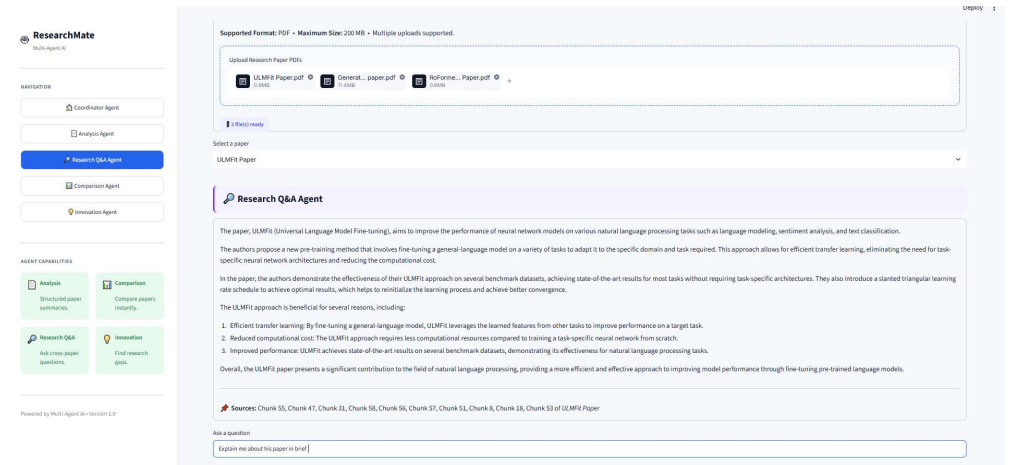


Figure: Q&A Agent citing exact source chunks used to ground its answer.

Comparison & Innovation Agents

Cross-paper intelligence: side-by-side comparison tables, and AI-surfaced research gaps.

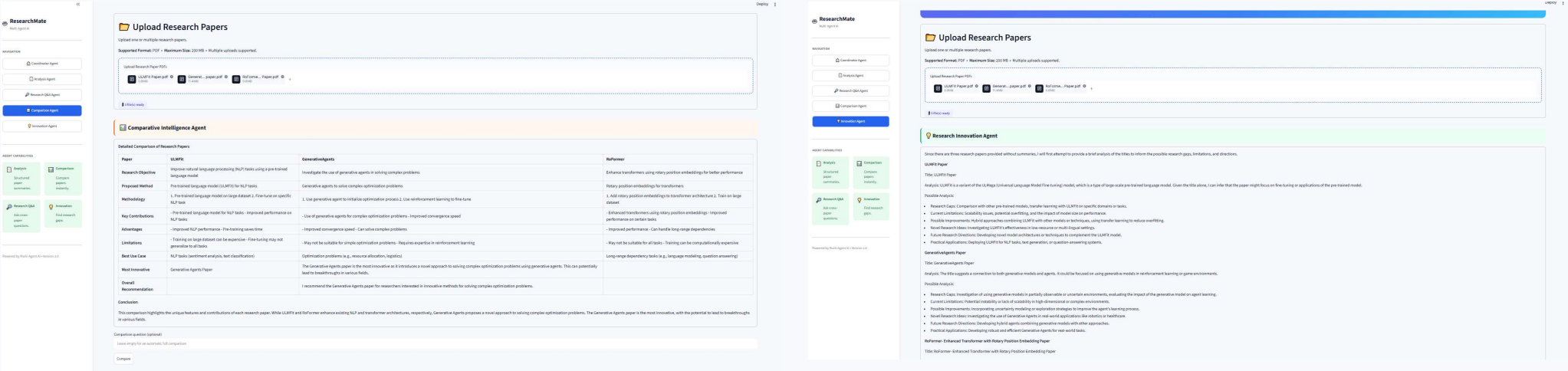


Figure: Comparative Intelligence Agent — auto-generated markdown table across 3 papers.

Figure: Research Innovation Agent — research gaps and future directions.

Comparison Agent: compares 2+ papers on objective, method, contributions, advantages, limitations, best use case, and recommends the most innovative paper.

Innovation Agent: adapts its analysis for single vs. multiple papers — surfacing shared gaps, hybrid ideas, and a concrete future-research proposal.

Technology Stack

Built on open-source NLP and vector-search libraries with a free-tier LLM backend.

Layer	Technology	Purpose
Frontend	Streamlit	Interactive multi-page UI with gradient theme
Chunking	LangChain (RecursiveCharacterTextSplitter)	1000-char chunks, 200-char overlap
Embeddings	Sentence-Transformers (all-MiniLM-L6-v2)	Dense semantic vectors for chunks & queries
Vector DB	FAISS (IndexFlatL2)	Fast nearest-neighbour similarity search
LLM	Groq API (llama-3.1-8b-instant)	Free-tier inference for all 4 specialist agents
PDF Parsing	PyMuPDF (fitz)	Raw text extraction from uploaded PDFs

System Demonstration

Sidebar-driven navigation, ChatGPT-style coordinator chat, and colour-coded agent output panels.

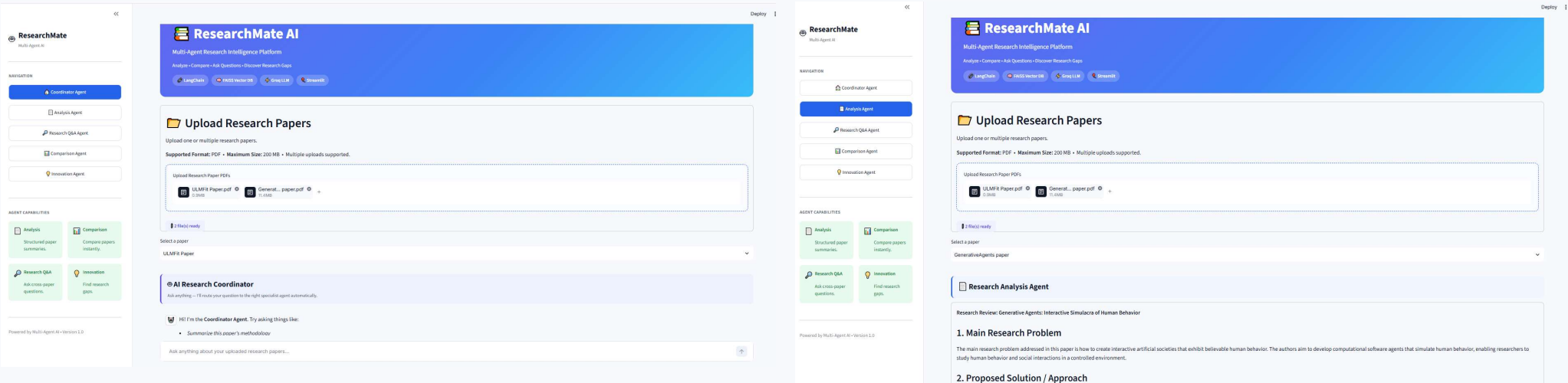


Figure: Coordinator Agent chat — natural-language questions auto-routed to specialists.

Figure: Research Analysis Agent — structured, color-coded 7-point review.

Interface highlights: gradient hero header · sidebar with active-page highlighting · agent capability cards · upload cap of 6 papers with graceful warning · fixed toast-spam bug for repeat uploads.

Challenges & Outcomes

Key Challenges

- Groq free-tier rate limits (6,000 tokens/min) — solved with graceful error handling in llm.py
- Balancing chunk size vs. retrieval precision (settled on 1000 chars / 200 overlap)
- Streamlit session-state re-runs causing duplicate processing on repeat uploads
- Designing keyword-routing robust enough for combined-intent queries
- Keeping the UI modular (agents / ui / utils / services) as features grew



PDF Pipeline

Multi-paper ingestion & chunking, fully automated



Semantic Search

FAISS retrieval operational and fast



5 Working Agents

Coordinator + 4 specialists, all live



Polished UI

Gradient theme, sidebar nav, chat interface

What's Next for ResearchMate

A working prototype today — with a clear roadmap to a production-grade research companion.



Smarter Retrieval

Cross-encoder re-ranking on top of FAISS for even more precise, page-accurate answers.



Persistent Workspaces

Save vector indexes across sessions so past papers stay searchable instantly.



One-Click Literature Review

Auto-export a full, citation-backed literature review draft from all four agents.

CONCLUSION

ResearchMate is a complete, working multi-agent RAG system — from PDF ingestion to grounded, cited answers — built around how researchers actually work: reading, comparing, questioning, and ideating.

5

AI Agents

4

RAG Pipeline Stages

100%

Grounded, Cited Answers

Thank You

ResearchMate: Multi-Paper Analyzer · Python · Streamlit · FAISS · Sentence-Transformers · Groq LLM